

Skills Summary

Rewriting Expressions with Distributive Structure

Use this reference while completing your worksheet or when studying.

Skill 1 — Distributing a factor

Rule: $a(b + c) = ab + ac$. The outside factor meets *every* term inside, signs included. In a measurement problem this is “ a copies of the whole thing”, so the unit of the answer is the unit of the bracket.

Example: A path is made of 6 sections, each $(2x + 7)$ m long. Write the total length in metres without brackets.

Step 1. Six copies of one length is a multiplication, and the length has two parts, so the 6 has to reach both of them.

Step 2. $6 \cdot 2x = 12x$ and $6 \cdot 7 = 42$.

$\Rightarrow$ **$12x + 42$** metres

Try it: Four sections, each $(3x - 5)$ m long: write the total length in metres without brackets.

check: $4(3x - 5)$

Skill 2 — Taking a common factor out

Rule: $ab + ac = a(b + c)$ — the same statement read right to left. Take the largest number that divides every coefficient, and the lowest power of the variable that appears in *every* term. Check by re-distributing.

Example: A ribbon of length $(10x + 35)$ cm is to be cut into equal pieces. Write the length as a product.

Step 1. I want to see how many equal pieces the length splits into, so I look for the largest number that divides both coefficients.

Step 2. 10 and 35 share 5, and $10x \div 5 = 2x$, $35 \div 5 = 7$.

$\Rightarrow$ **$5(2x + 7)$** — five pieces of $(2x + 7)$ cm.

Try it: Write $8x^2 + 12x$ as a product of its greatest common monomial factor and one remaining factor.

check: $(8 + 12x)(2x)$

Skill 3 — Choosing the form that answers the question

Rule: the two forms are equal but they show different things. *Factored* form shows what the quantity is divisible by — how many equal parts, what the per-item rate is. *Expanded* form shows the fixed part and the variable part as separate terms. Rewrite into whichever one puts the asked-for quantity on the page.

Example: A walkway $(5x + 20)$ m long is split into 5 equal sections. How long is one section, in metres?

Step 1. The question is about equal parts, so the factored form is the one that will show the answer without dividing term by term.

Step 2. $5x + 20 = 5(x + 4)$, which reads as 5 copies of $(x + 4)$ metres.

Step 3. One section is the second factor.

$\Rightarrow x + 4$ metres

Try it: A walkway $(7x + 21)$ m long is split into 7 equal sections. How long is one section, in metres?

check $x + 3$

(!) Watch out: a minus sign in front of a bracket is a factor of -1 and reaches every term: $40x - 5(3x + 20) = 40x - 15x - 100$, not $40x - 15x + 100$.